**Speciation Study Guide**

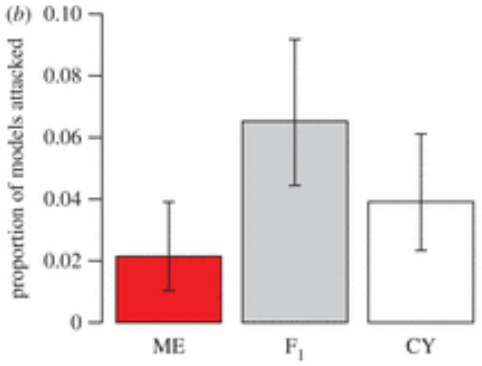
This study guide provides examples for the topics that you might see on the exam over speciation and reproductive barriers. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and your homework questions to review all the topics that have been covered.

1. Define each of the species concepts provided below. What are the pros and cons for each species concept? For what organisms does each species concept work well? For what organisms does each species concept **not** work well?
    a. Biological Species Concept
    b. Morphological Species Concept
    c. Phylogenetic/Lineage Species Concept

2. Rank these concepts in order of most general to most specific: a) gametic isolation, b) reproductive isolating mechanism, c) sperm-egg incompatibility in sea urchins, d) prezygotic isolating mechanism

3. For each of the types of speciation listed below, describe the general pattern of how speciation occurs and provide an example of speciation that has followed this pattern
    a. Allopatry through vicariance
    b. Allopatry through the founder effect
    c. Sympatry

4. The figure below shows the proportion of individuals attacked by birds in a population of butterflies in central Panama. The red bar represents *Heliconius melpomene* (labeled ME), the white bar represents *Heliconius cydno* (labeled CY), and the gray bar represents F1 hybrids between the two. Do the data in the figure indicate a reproductive barrier between these two species? If so, what kind of barrier is suggested?




5. For each of the scenarios below, **describe what type of reproductive barrier might be present and whether this barrier is pre-zygotic or post-zygotic**

    a. Two species of frogs belonging to the same genus occasionally mate, but the embryos stop developing after a day and then die.

    b. Rocky Mountain juniper (*Juniperus scopulorum*) and one-seeded juniper (*J. monosperma*) have overlapping ranges. Pollen grains (which contain sperm cells) from one species are unable to germinate and make pollen tubes on female ovules (which contain egg cells) of the other species

    c. Beetle pollinators of a particular plant are attracted to its flowers' bright orange color. The beetles not only pollinate the flowers, but they mate while inside of the flowers. A mutant version of the plant with red flowers becomes more common with the passage of time. A particular variant of the beetle prefers the red flowers to the orange flowers. Over time, these two beetle variants diverge from each other to such an extent that interbreeding is no longer possible.

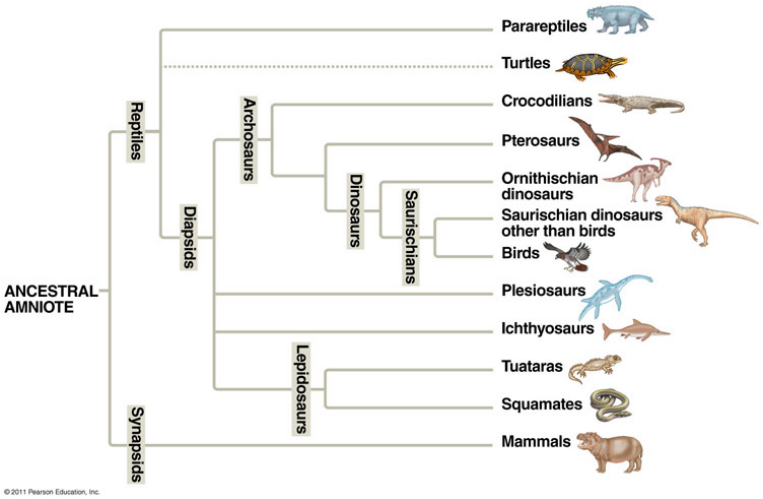
d. Three populations of crickets look very similar, but the males have courtship songs that sound very different. Females of each population prefer the song of the males in their population.

e. Many songbirds breed in North America in the spring and summer and then migrate to Central and South America in the fall. They spend the winter in these warmer areas, where they feed and prepare for the spring migration north and another breeding season. Two hypothetical species of sparrow, A and B, overwinter together in mixed flocks in Costa Rica. In spring, species A goes to the east coast of North America, and species B goes to the west coast.

f. On the Bahamian island of Andros, mosquitofish populations live in various, now-isolated, freshwater ponds that were once united. Currently, some predator-rich ponds have mosquitofish that can swim in short, fast bursts; other predator-poor ponds have mosquitofish that can swim continuously for a long time. When placed together in the same body of water, the two kinds of female mosquitofish exhibit exclusively breed with males that have the same swimming phenotype as the female.

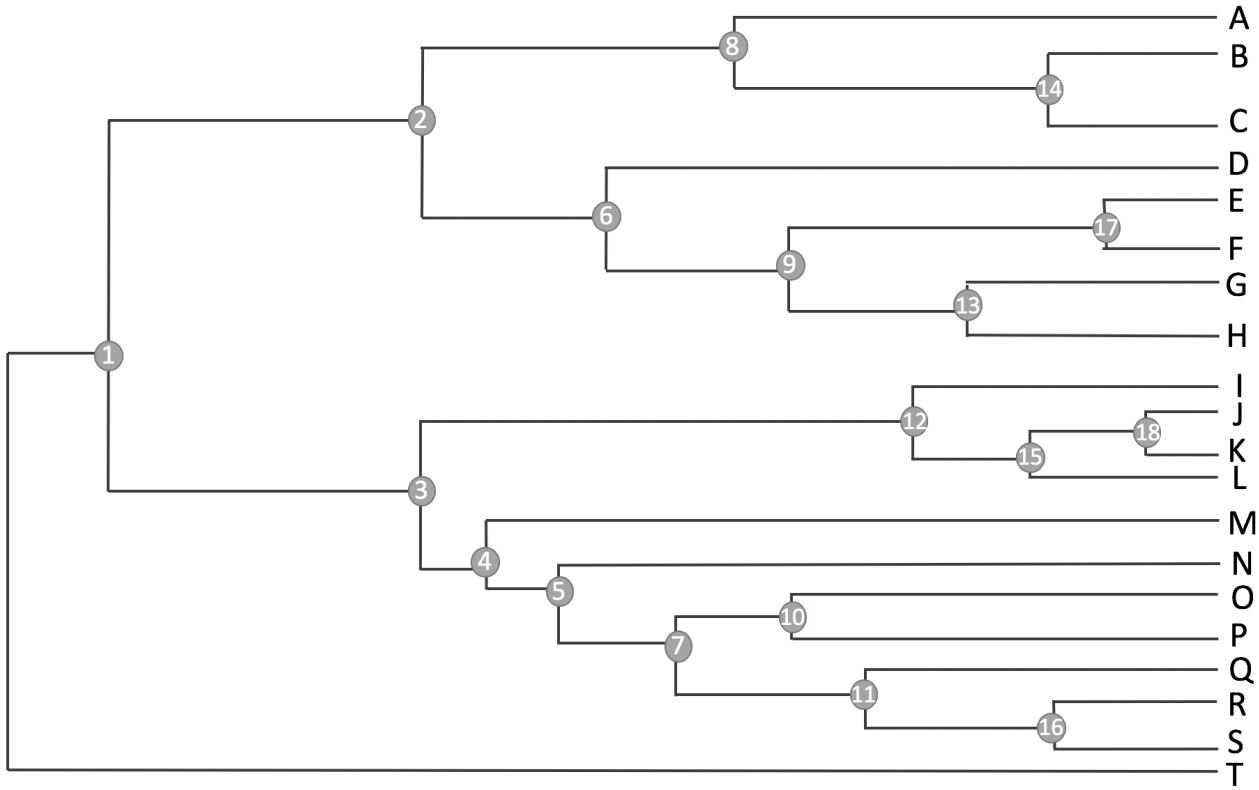
6. For these questions, refer to the phylogenetic tree at right.
   a. What other reptile is the closest **living** relative of birds?

   b. Why is the line for turtles dotted instead of a solid line like the others?




7. What is the difference between homologous traits and analogous traits? Determine whether each of the traits listed below are homologous or analogous traits.
   a. Limb bones of humans and cats
   b. Wings of birds and insects
   c. Streamlined body shape in dolphins and sharks
   d. Cactus spines and conifer needles

8. Below is a fictional phylogenetic tree showing taxa (in letters) and their common ancestors (numbered branching points). Use the tree to answer the questions below.




a. For each of the groups of taxa below, identify the numbered branch point that represents their most recent common ancestor
   i. Taxa A, B, and C
   ii. Taxa H, I, and J
   iii. Taxa N and O
   iv. Taxa P, Q, and R
b. For each of the groups of taxa below, determine if the group is or is not monophyletic. If the group is **NOT** monophyletic, list the taxa that would need to be included in order to make the group monophyletic.
   i. Taxa A, B, and C
   ii. Taxa H, I, and J
   iii. Taxa N and O
   iv. Taxa P, Q, and R
c. If a phenotype is shared by the following groups, would you consider is analogous or homologous?
   i. Taxa C, K, T
   ii. Taxa I, J, K, L
   iii. Taxa S, R, Q, and O